Goodness-of-fit Diagnostics for State Space Models in Neuroscience

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Abstract

State space models are widely used in neuroscience to relate neural activity to latent dynamic brain states. This paper introduces diagnostics (KL divergence and HPD overlap) to assess model goodness-of-fit by examining consistency between posterior distributions and component likelihood distributions.

Keywords: state space models, goodness-of-fit, model diagnostics, Bayesian inference, neuroscience, neural decoding

1 Introduction

2 Methods

2.1 State Space Model Framework

2.2 Diagnostic Metrics

3 Results

3.1 Simulated Data

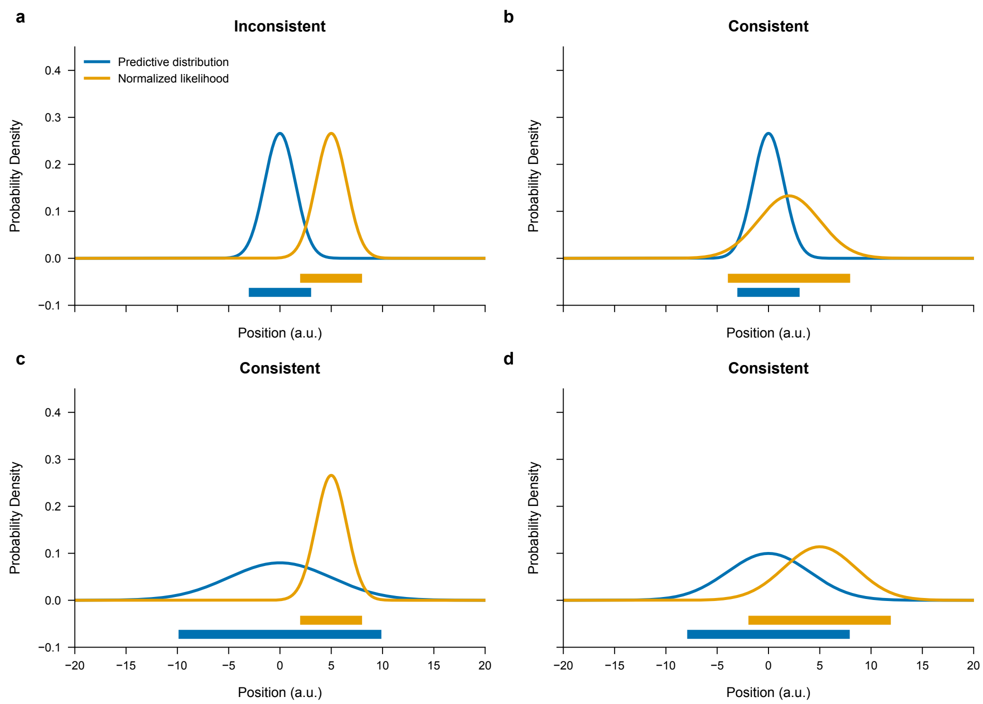

Figure 1: **Distribution consistency examples.** Four scenarios showing predictive distribution and normalized likelihood with varying degrees of consistency.

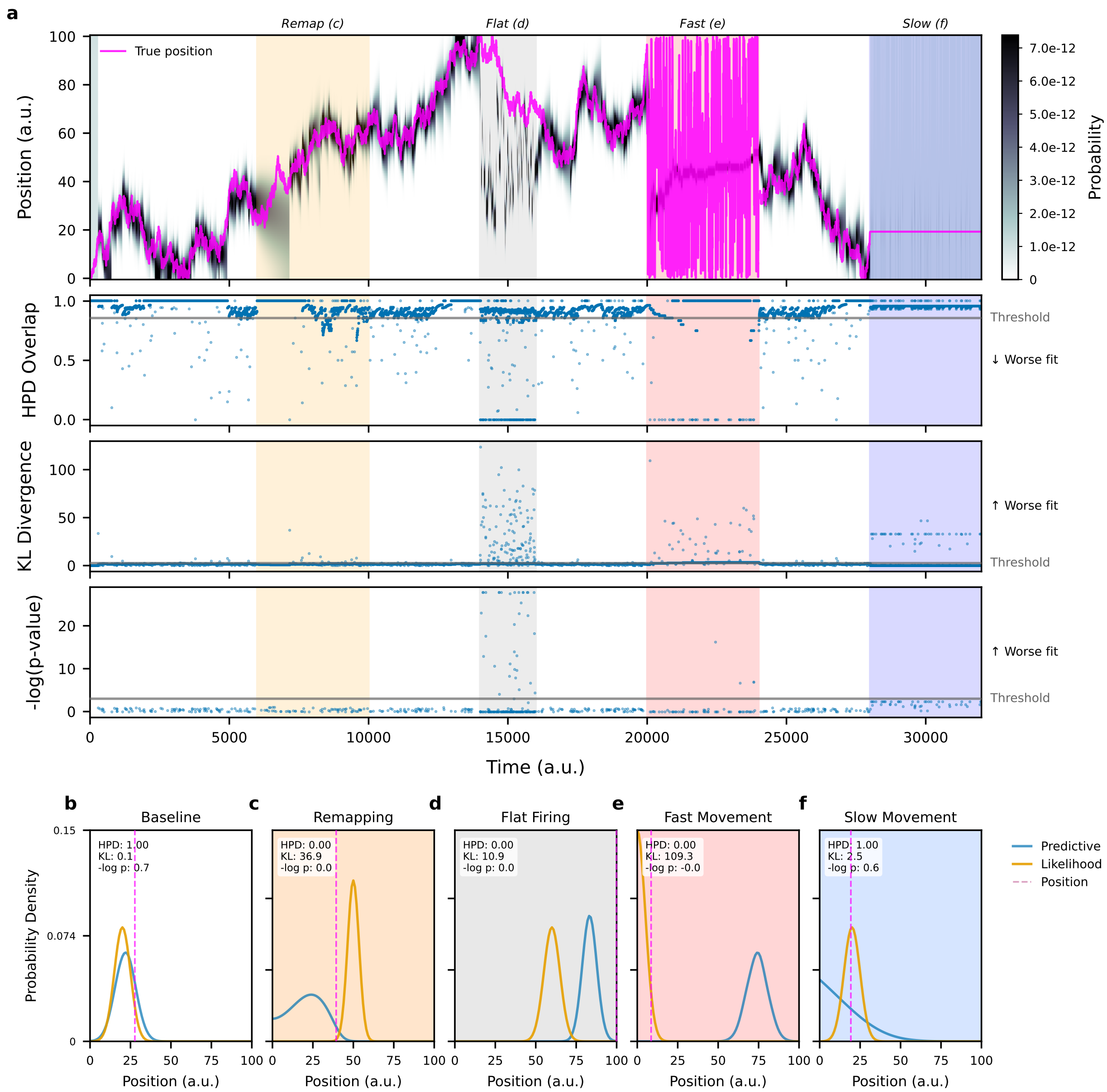

Figure 2: **Diagnostic demonstration on simulated data.** Time-resolved diagnostics showing periods of good and poor model fit.

4 Discussion

5 Conclusion

Code and Data Availability

All code to reproduce figures and analyses is available at https://github.com/Eden-Kramer-Lab/statespacecheck-paper. The `statespacecheck` Python package is available at https://github.com/Eden-Kramer-Lab/statespacecheck.

Author Contributions

SZ, UTE, and ELD wrote the manuscript.

Competing Interests

The authors declare no competing interests.

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